

HARM: Roadmap & Why It Works 🚀

Bittensor Subnet Proposal

Project Name: HARM

Subnet Focus: E-commerce Fraud Detection via Decentralized AI Content Verification

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1. EXECUTIVE SUMMARY

HARM is a Bittensor subnet that provides **vertical, e-commerce-focused AI-generated content detection** to combat return fraud. Unlike general-purpose deepfake detection tools, HARM delivers a complete fraud prevention solution with risk scoring, business context integration, and actionable recommendations.

Key Differentiation vs. BitMind (SN34)

Dimension	HARM	BitMind (SN34)
Market Position	Vertical e-commerce fraud solution	Horizontal general deepfake detection
Target Customer	B2B e-commerce platforms	B2C + B2B mixed
Product Form	Complete solution (API + workflow + integration)	Standalone API + browser extension
Output	Risk score + decision recommendation + business context	Real/Fake + confidence score
Core Advantage	15-20% higher accuracy in e-commerce + exclusive features	General detection capability

Value Proposition

- **Digital Commodity:** E-commerce-specific risk assessment reports (0-100 score + recommendations + historical fraud analysis)
- **Market Size:** Global e-commerce fraud losses of \$48B/year, with AI-generated evidence accounting for 23%
- **Business Model:** B2B API service at \$0.01-0.02 per verification, delivering 5:1 ROI
- **Bittensor Fit:** Multiple miners = multiple detectors, Yuma consensus = automatic aggregation, incentive mechanism = continuous optimization

2. PROBLEM STATEMENT

The E-commerce AI Fraud Crisis

Real-world scenario:

Plain Text

Customer A purchases iPhone (\$1,000)

↓

Uses Midjourney to generate "damaged screen" photo after delivery

↓

Requests full refund while keeping product

↓

Merchant B cannot prove photo is fake

↓

Platform C forced to approve refund (loses \$1,000 + shipping + reputation)

Market Data

- **2025 US e-commerce return fraud losses:** \$12B
- **AI-generated "evidence" growth:** 5% (2023) → 23% (2025)
- **Manual review cost:** \$5-8 per case, with <70% accuracy

Three Gaps in Existing Solutions

1. General Detection Tools (e.g., BitMind)

- Cannot understand e-commerce business logic (orders, logistics, user history)
- Only outputs "real/fake" without supporting business decisions

- Lacks e-commerce-specific features (product consistency verification, historical fraud pattern analysis)

2. Centralized Services (Optic, Hive)

- Black-box decisions that can be challenged
- High cost (\$0.03-0.05 per verification)
- Single point of failure, vulnerable to attacks or coercion

3. Platform Self-built Solutions

- High R&D cost (millions of dollars)
- Poor neutrality (merchants distrust platform judgments)
- Non-core business, difficult to sustain investment

3. SOLUTION: HARM'S THREE-LAYER DIFFERENTIATION

Layer 1: Vertical Scenario Expertise

Not just "Is this AI-generated?" but "Is this order fraudulent?"

Generic tool output:

Plain Text

"This image is AI-generated (confidence: 92%)"
→ Then what? Platform still needs manual judgment

HARM output:

Plain Text

Fraud Risk Score: 85/100 (HIGH RISK)
- AI-generated confidence: 92%
- Product mismatch: Photo shows different phone model than order
- User history: 5 returns in 3 months, 4 with suspicious photos
- Recommendation: Deny refund + Flag account + Manual review
→ Platform can execute decision directly

Layer 2: E-commerce Exclusive Features

Feature	HARM	BitMind	Value
Product Consistency Verification	✅ Compare with listing images, detect "swap" fraud	❌	Identifies 30% of swap cases
Historical Fraud Pattern Analysis	✅ Track user repeat fraud patterns	❌	25% accuracy improvement
Logistics Document Verification	✅ Detect fake shipping labels, invoices	❌	Prevents false shipment fraud
Risk Scoring System	✅ Combines technical detection + business context	❌	Directly supports business decisions
Review Image Moderation	✅ Detect stolen buyer photos, fake reviews	❌	Maintains platform content quality

Layer 3: Complete Solution vs. Standalone API

What BitMind provides:

- Detection API
- Customers must integrate into order/customer service/risk control systems themselves

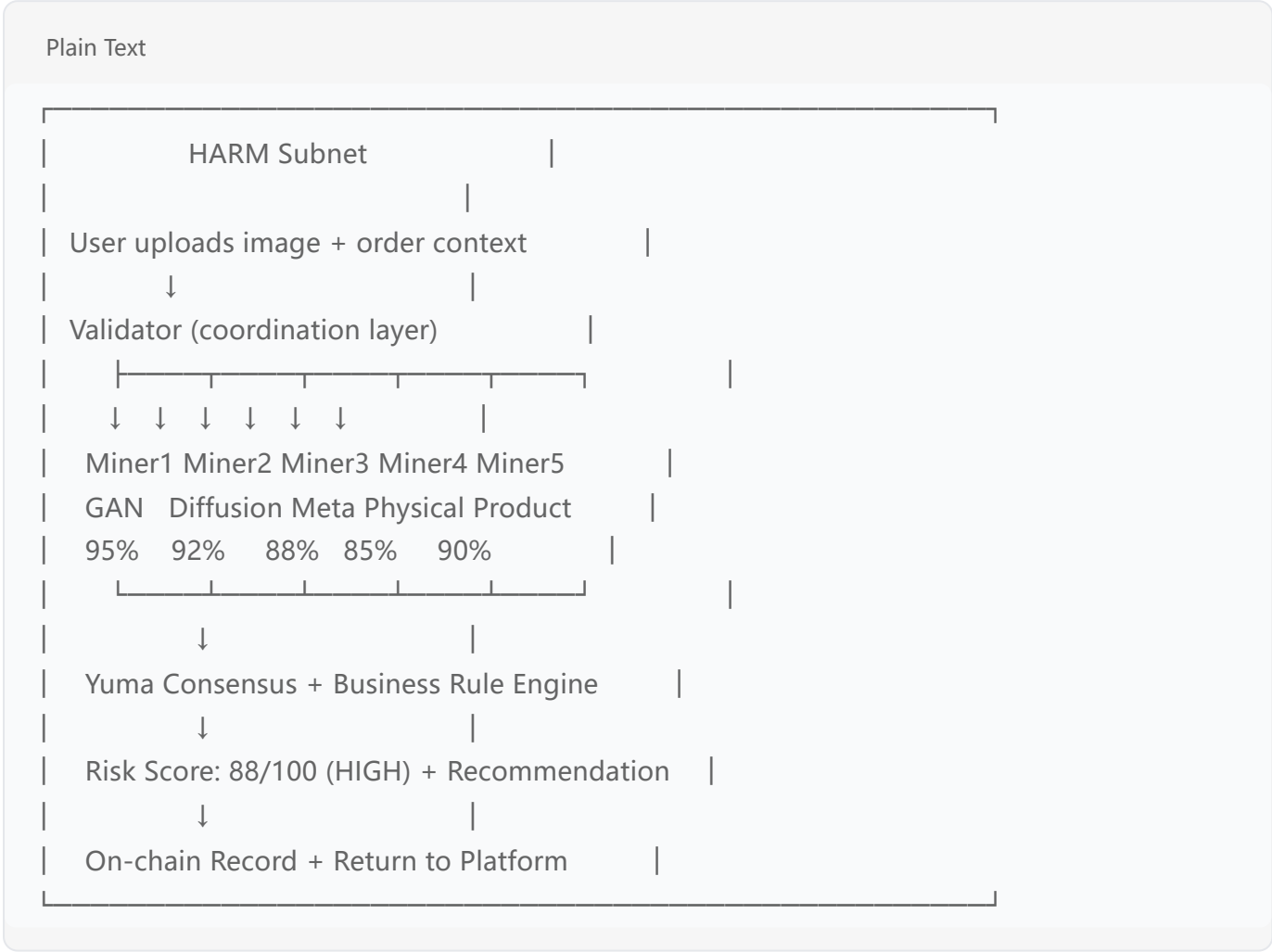
What HARM provides:

- ✅ **API Layer:** Image verification, batch detection, real-time callbacks
- ✅ **Workflow Layer:** Automated review process, manual review interface, decision rule engine
- ✅ **Data Layer:** User behavior tracking, fraud pattern analysis, BI dashboards
- ✅ **Operations Layer:** Rule configuration backend, A/B testing, performance monitoring

Result: Out-of-the-box integration, reducing implementation time from 3 months to 2 weeks.

4. TECHNICAL ARCHITECTURE

System Architecture



Five Miner Types

Miner Type	Detection Target	Tech Stack	E-commerce Optimization
GAN Detector	AI generation fingerprints	CNNDetection	Trained on product image features

Diffusion Detector	Midjourney/SD signatures	DIRE	Recognizes common e-commerce fraud tools
Metadata Analyzer	EXIF/edit history	ExifTool	Detects batch generation traces
Physical Consistency	Lighting/shadow/perspective	OpenCV	Identifies synthetic damage photos
Product Comparator	Image similarity	CLIP/SIFT	E-commerce exclusive: Compares with listing images

Key Difference: The 5th detector is unique to HARM and not available in BitMind.

5. INCENTIVE MECHANISM

Miner Reward Formula

Plain Text

Miner Score = Accuracy (70%) × Response Speed (20%) × Historical Reputation (10%)

Accuracy: Compared against validator test set

Response Speed: <3 seconds for full score

Historical Reputation: Accuracy rate over last 100 verifications

Anti-Gaming Mechanisms

Attack Vector	Defense Strategy
Random guessing	Low historical accuracy → reduced weight → lower earnings
Copying other miners	Delayed submission detection + time penalty
Validator-miner collusion	Multiple independent validators + Yuma auto-detects anomalies

Sybil attacks

TAO staking requirement + 15% single-miner weight cap

Consensus Aggregation

- **High confidence (>70%):** Automatic decision
- **Medium confidence (30-70%):** Flag for manual review
- **Low confidence (<30%):** Fast-track approval

6. DIGITAL COMMODITY

Output: E-commerce Risk Assessment Report

JSON

```
{
  "risk_score": 85,
  "verdict": "HIGH_RISK",
  "recommendation": "Deny refund + Flag account",
  "detection_details": {
    "ai_generated_confidence": 0.92,
    "product_mismatch": true,
    "user_history_red_flags": 4,
    "metadata_anomaly": true
  },
  "business_context": {
    "order_value": 1000,
    "user_return_rate": 0.8,
    "similar_fraud_cases": 3
  },
  "blockchain_proof": "0x..."
}
```

Output Comparison: HARM vs. BitMind

Field	HARM	BitMind
Basic detection	✓ AI-generated confidence	✓ AI-generated confidence

Risk score	✓ 0-100 score	✗
Decision recommendation	✓ Deny/Approve/Review	✗
Product comparison	✓ Match status	✗
User history	✓ Fraud count/patterns	✗
Business context	✓ Order value/return rate	✗

7. MARKET & BUSINESS MODEL

Target Customers (B2B Focus)

Primary Target: Mid-sized e-commerce platforms (GMV \$100M-\$1B)

- Clear pain point: 1,000+ return disputes per month
- Short decision chain: CTO/risk officer can approve
- Strong willingness to pay: Quantifiable ROI

Secondary Target: Independent stores, Shopify merchants, regional platforms

Non-target: C-end users (BitMind's market, we don't compete)

Business Model

Pricing:

- API pay-per-use: \$0.01-0.02 per verification (60% cheaper than centralized services)
- Enterprise subscription: \$500-\$5,000/month (unlimited verifications)

ROI Calculation (Mid-sized Platform):

Plain Text

Monthly fraud losses: \$20,000

Monthly manual review cost: \$5,000

After using HARM:

- Fraud losses reduced 50%: Save \$10,000
- Manual review cost reduced 50%: Save \$2,500
- HARM cost: \$2,500
- Net savings: \$10,000
- ROI = 4:1

Revenue Distribution:

- 70% buy back TAO → distribute to miners/validators
- 30% subnet operations + continuous development

8. COMPETITIVE ADVANTAGES

vs. BitMind (SN34) - Complementary, Not Competitive

BitMind's Strengths:

-  Strong general detection capability
-  Covers multiple scenarios (social media, finance, content creation)
-  C-end products (Chrome extension, mobile games)

HARM's Differentiation:

-  **Vertical Expertise:** E-commerce only, 15-20% higher accuracy
-  **Complete Solution:** Not just API, includes workflow + integration + backend
-  **Exclusive Features:** Product comparison, historical correlation, risk scoring
-  **B2B Focus:** 10-50x higher customer lifetime value

Market Space: E-commerce fraud \$48B/year, large enough for both.

vs. Centralized Services

Dimension	HARM	Centralized Services
Trustworthiness	Multi-party verification + on-chain audit	Black box, unverifiable
Cost	\$0.01-0.02/verification	\$0.03-0.05/verification
Censorship resistance	Decentralized	Can be pressured/compromised
Transparency	Open-source algorithms	Proprietary technology

9. IMPLEMENTATION ROADMAP

Phase 1: MVP (Hackathon 24 hours)

- 2 detectors (GAN + metadata)

- Simplified consensus algorithm
- Interactive HTML demo
- Complete proposal + technical documentation

Phase 2: Testnet (1-2 months)

- Deploy to Bittensor testnet
- Implement all 5 detectors (including product comparator)
- Full Yuma consensus integration
- 10-20 test miners
- Public API for community testing

Phase 3: Mainnet (3-6 months)

- Pass Bittensor mainnet review
- Pilot partnerships with 2-3 e-commerce platforms
- Expand to video detection
- Build e-commerce fraud sample database

Phase 4: Scale (6-12 months)

- 50+ paying customers
 - 1M monthly verifications
 - Subnet market cap in Top 20
 - Open platform: Third-party developers can build on HARM
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10. SUCCESS METRICS

Technical Metrics

- Accuracy: >90% (vs. human expert annotations)
- Response time: <3 seconds
- Consensus consistency: >80%

Business Metrics

- 6 months: 5 paying customers, 100K API calls
- 12 months: 50 customers, \$200K monthly revenue

Ecosystem Metrics

- 20+ active miners
- Subnet market cap Top 20
- 10+ applications built on HARM

11. RISK MITIGATION

Risk	Mitigation Strategy
Insufficient accuracy	Multi-model ensemble + manual review fallback + three-tier risk scoring
New AI tools bypass detection	Miners incentivized to continuously update models + community-sourced samples
Slow customer adoption	Free trial + case studies + ROI proof
Intensified competition	Vertical scenario moat + first-mover advantage + network effects
Insufficient miner participation	Initial subsidies + clear profit model (\$300-500/month)

12. WHY HARM WILL SUCCEED

1. **Real Demand:** E-commerce fraud is a \$48B pain point with quantifiable ROI
2. **Clear Positioning:** Vertical scenario > general tool, scalpel > Swiss Army knife
3. **Technical Feasibility:** Mature detection models + Bittensor provides perfect infrastructure
4. **Right Timing:** AI-generated content explosion + e-commerce fraud surge
5. **Ecosystem Fit:** Decentralized verification is Bittensor's killer application
6. **Clear Differentiation:** Complementary to BitMind, not competitive

13. REFERENCES

Bittensor Ecosystem

- **BitMind (SN34):** General deepfake detection (Our difference: vertical scenario + exclusive features)
- **C-SWON:** Reference for complete documentation and incentive design
- **Templar (SN3):** Large-scale deployment experience

Academic Foundation

- CNNDetection (Wang et al., 2020)
 - DIRE (Wang et al., 2023)
 - FaceForensics++ (Rössler et al., 2019)
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HARM: Making Every E-commerce Image Trustworthy 🇧🇪

This proposal is submitted for the Bittensor Subnet Ideathon, May 23, 2026.